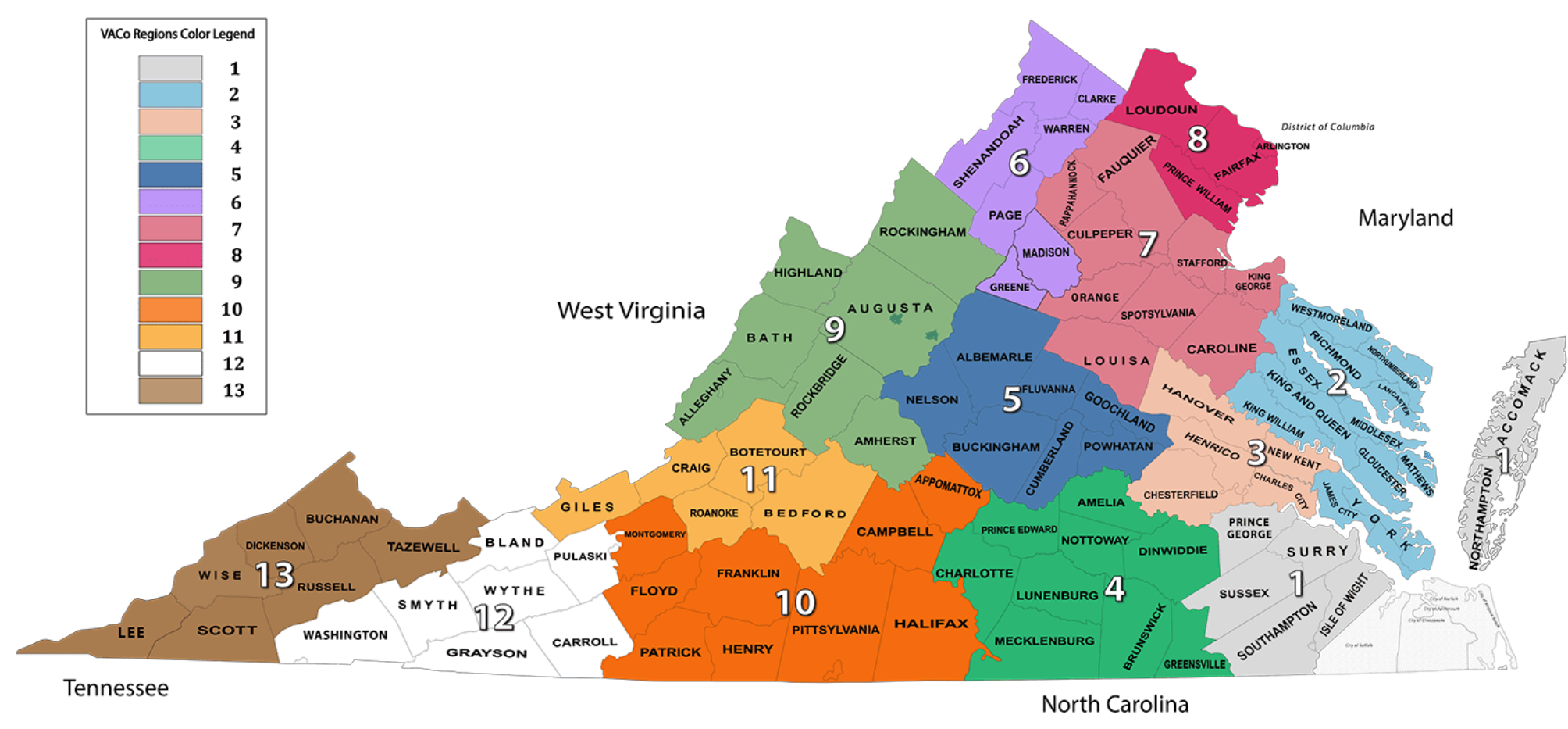




Analysis of COVID-19 Cases and 2020 Presidential Election Results in Virginia

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I. Background

The relevance of the ongoing COVID-19 pandemic led us to explore the distribution of COVID-19 cases across Virginia. For comparison, we also chose to analyze Virginia 2020 Presidential Election data by county. Our aim was to determine whether 2020 Election results were correlated with COVID-19 cases and, if so, which candidate corresponded to higher percentages of COVID-19 cases.

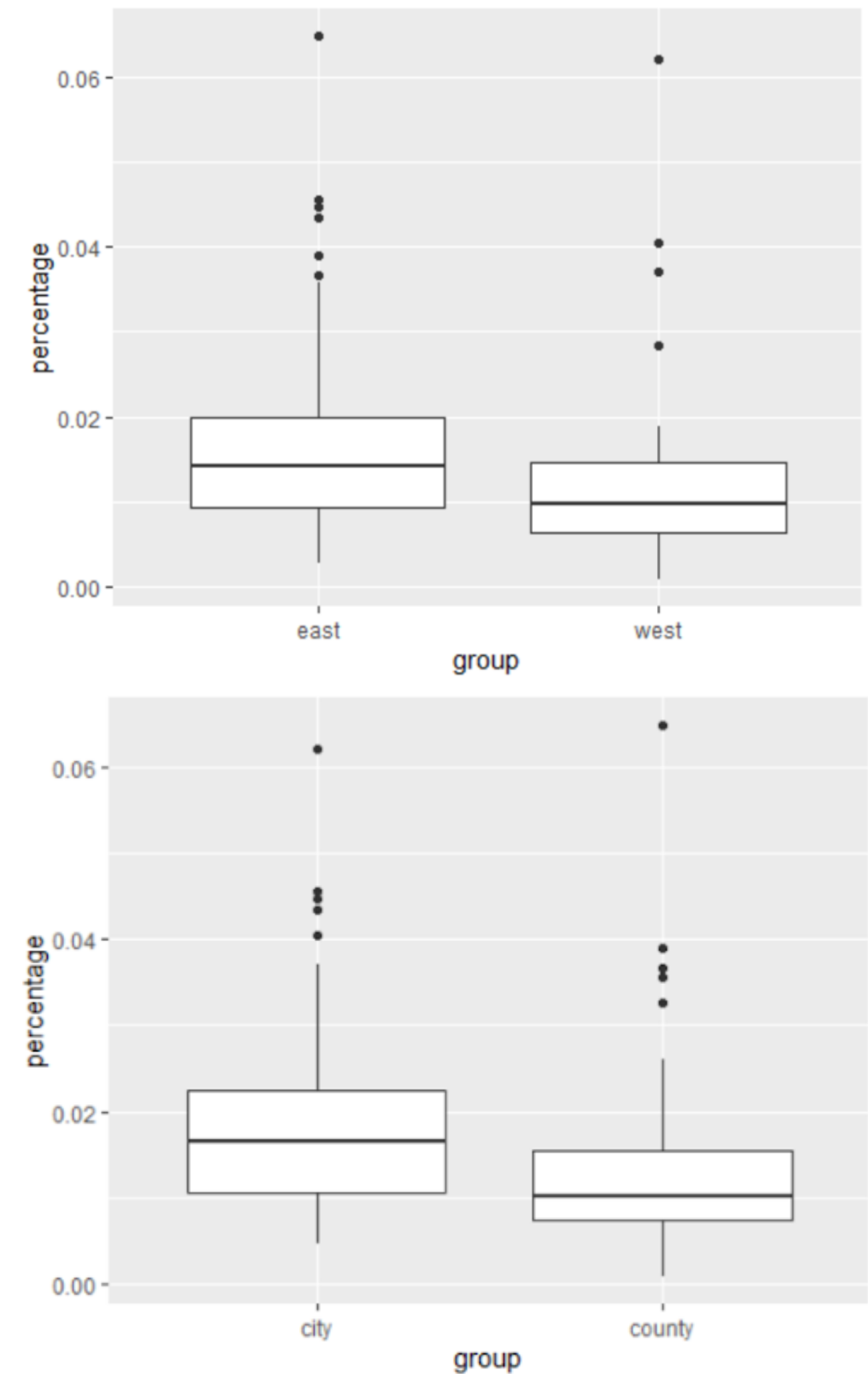
To test this question, we completed a variety of statistical techniques to test correlations, as well as analysis on both datasets separately.

We analyzed two main datasets: COVID-19 data, collected on September 14th, containing the cumulative case numbers in each county/city of Virginia, and 2020 election votes in each locality in Virginia containing the percentage of votes for Joe Biden (D) and percentage of votes for Donald Trump (R) by locality.

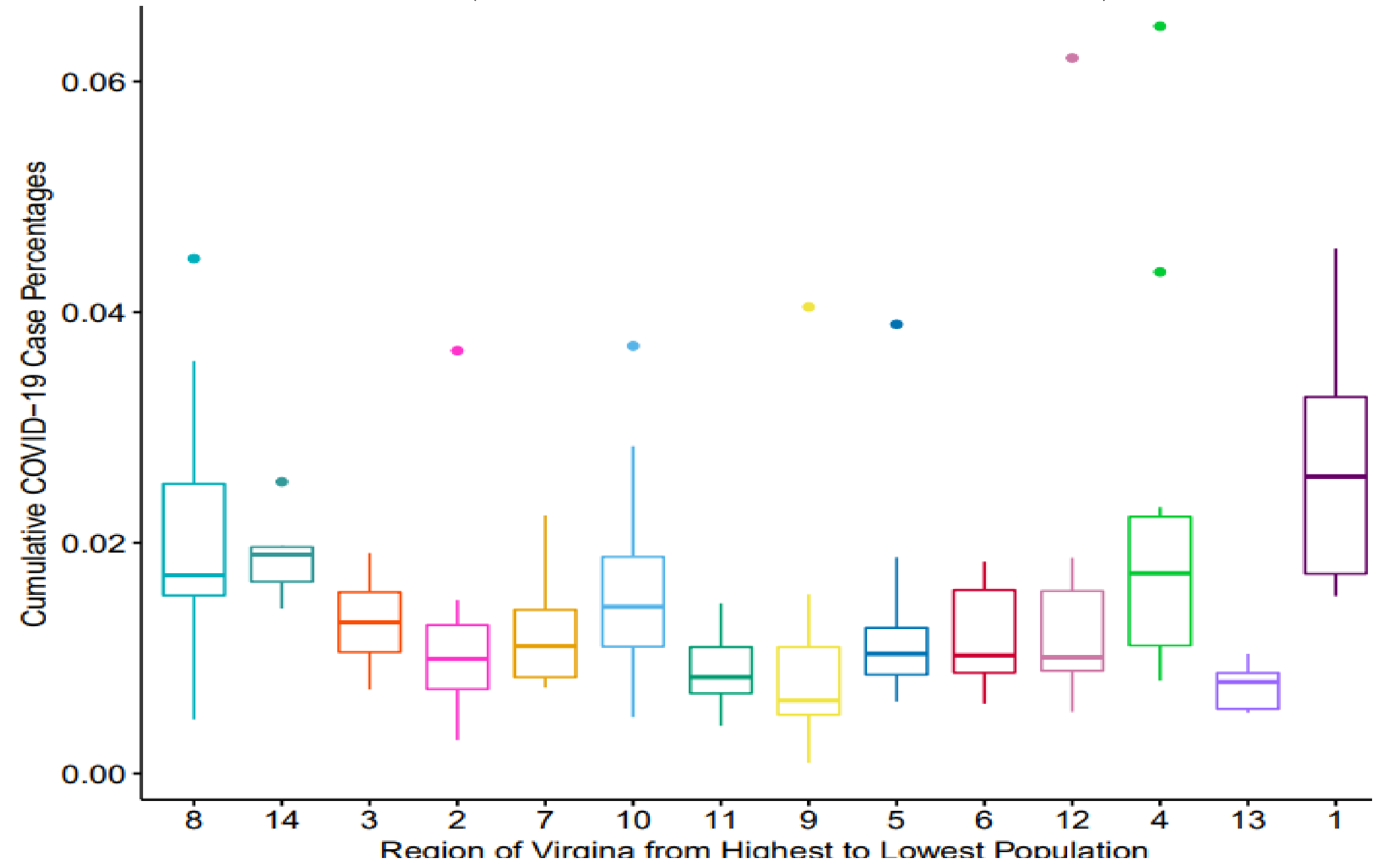
We also utilized the fourteen regions of Virginia as defined by the Virginia Association of Counties to organize our data. These regions can be seen in the image above..

II. COVID-19 Dataset Analysis by Region

The COVID-19 data includes county name, death total, and total cases in that county or city. We also created a new variable for the dataset using additional population data by calculating the percentage of COVID-19 cases for each locality (dividing the case total by the given locality's population size), and then sorted this dataset by the regions outlined in I.

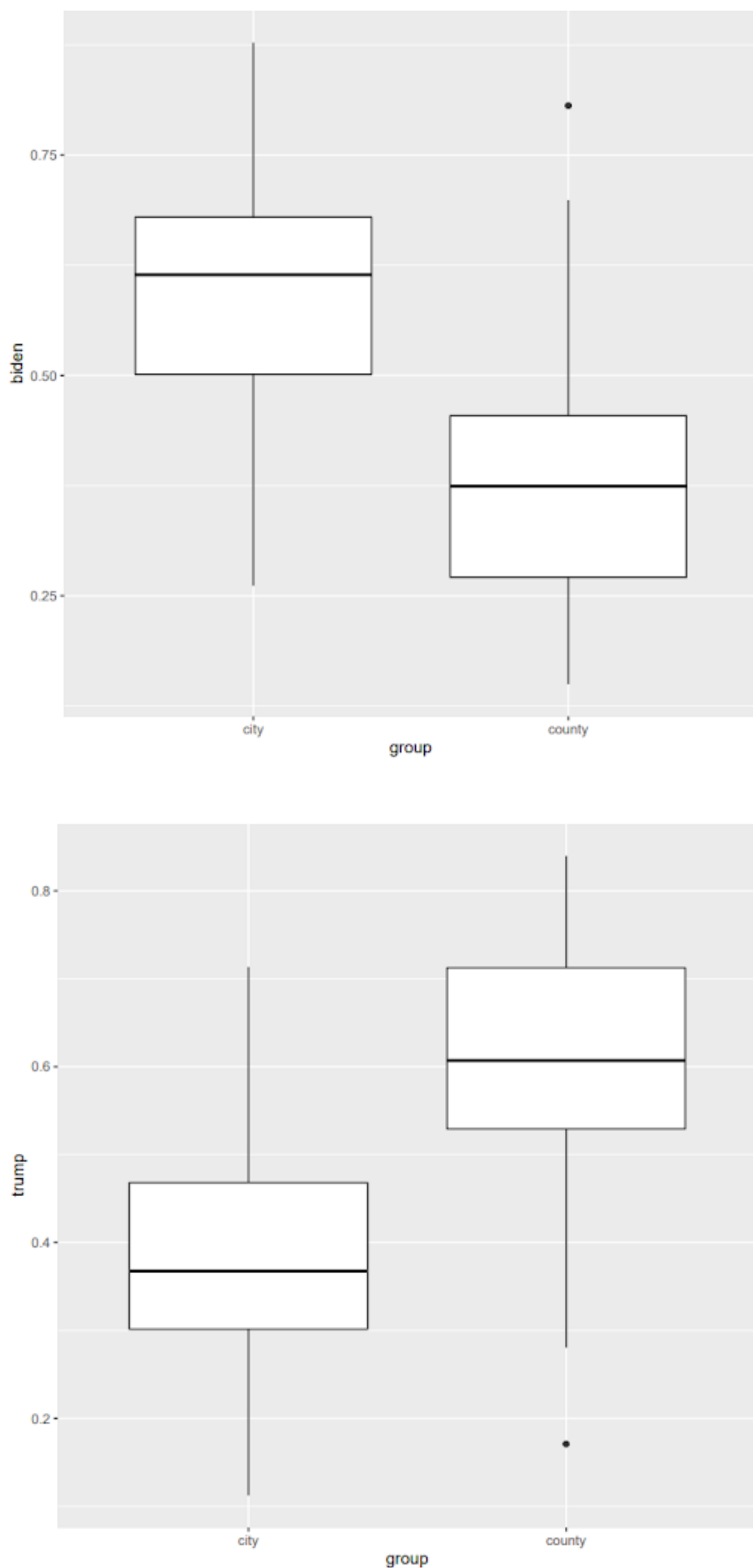


We generated several plots to help visualize this data, and to compare different regions and localities based on their affliction percentages. Through Kruskal-Wallis and Welch t-tests, we were able to statistically conclude that eastern counties had significantly higher case numbers than western counties (p-value = 0.2445 > 0.05), and that cities had significantly higher case numbers than counties (p-value = 0.008948 > 0.05).



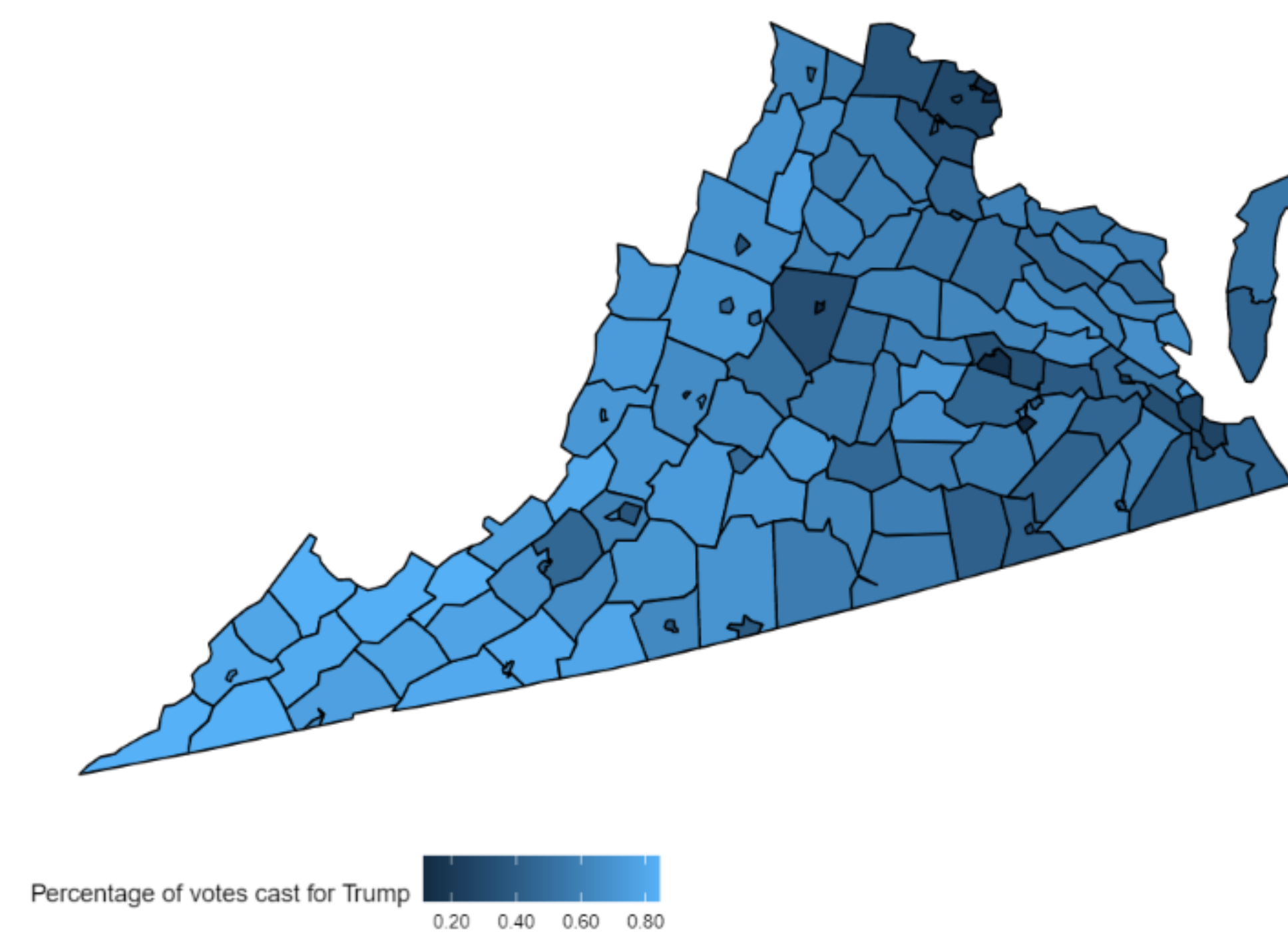
III. Election Data

In analyzing the 2020 Election Data, we found that counties voted for Trump (R) significantly more than cities. Conversely, cities were more likely to vote for Biden (D). We ran hypothesis tests to determine the significance of these results (p-value = 1.009e-08 < 0.05 for Biden). Below are plots for Biden and Trump votes, respectively, in cities (left) and counties (right).



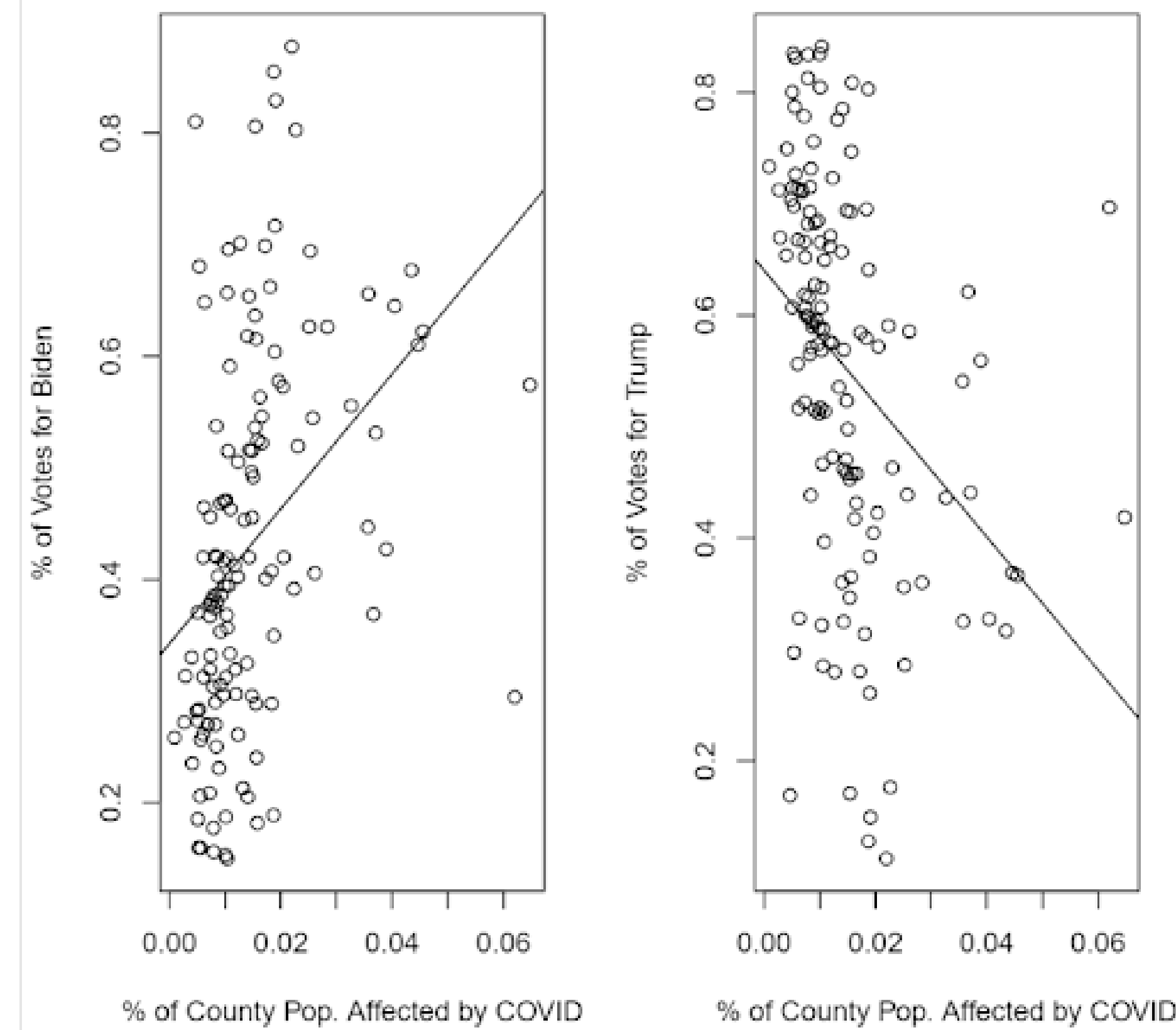
IV. Election and COVID-19 Dataset Comparison

A map of localities with Trump (R) votes (light blue) is shown below. It should be noted that, despite more localities voting for Trump (R), the Commonwealth of Virginia still went to Biden (D) due to the much larger populations of those localities.



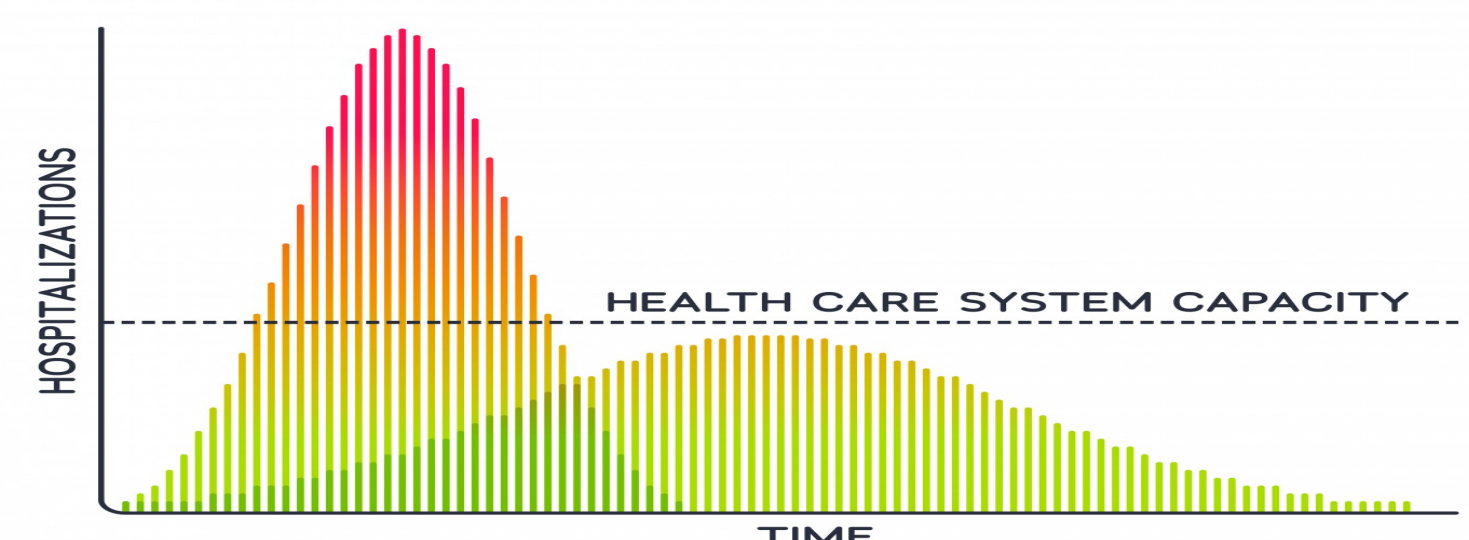
With the understanding that cities (with higher COVID-19 rates) tended to vote more for Biden and, likewise, counties with lower rates tended to vote for Trump, we wanted to test if this was statistically significant.

To determine if COVID-19 rates were correlated with voting patterns, Pearson correlation coefficients were calculated: Biden and COVID rates had a 0.3817212 correlation and Trump and COVID rates had a -0.3733086 correlation. Although the correlation isn't high, it does indicate that more people voted for Biden in higher COVID rate areas.



V. Moving Forward: Future Work

In future work, it may be interesting to compare the highlighted datasets with population densities in the studied regions, or with disease model predictions for each region. This may reveal a better vision of how COVID-19 affected voting patterns, and we are curious if other data would support the correlation we found.



Employed Datasets and Figures

I: "Cumulative COVID-19 Cases and Deaths by Virginia locality", Official Site of the Commonwealth of Virginia (Virginia.gov)

II: "The 2020 November General", Virginia Department of Elections

III: "Virginia Population Estimates", Demographics Research Group at UVA's Weldon Cooper Center

IV: "Virginia Association of Counties Regions", Virginia Association of Counties